SIEGEL'S THEOREM

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This worksheet works through Siegel's two proofs, from [Sil09, §§IX.3–IX.4], that an elliptic curve over a number field K has only finitely many K-rational points with coordinates in $\mathcal{O}_S$, although it can have infinitely many rational points.

Standing notation.

- A number field K with algebraic closure $\overline{K}$, set of places $\mathcal{P}_K$ and set of archimedean places $\mathcal{P}_K^\infty$.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v. If w is a place of a finite extension of K above v, then $|a|_w = |a|_v$ for $a \in K$.
- The height $H_K(a) \coloneqq \prod_{v\in\mathcal{P}_K} \max(1, |a|_v)^{n_v}$ of $a \in K$, which satisfies $H_K(a) \geqslant 1$ [Sil09, §VIII.5].
- A finite set $S \subseteq \mathcal{P}_K$ containing $\mathcal{P}_K^\infty$. The ring of S-integers $\mathcal{O}_S \coloneqq \{a \in K : |a|_v \leqslant 1 \text{ for all } v \notin S\}$ and its group of units $\mathcal{O}_S^\times$, the group of S-units.

How to use this. Allow about 20 minutes for Problem 1.4 and 40 for Problem 2.3. The two problems are independent. In a first pass, Problem 2.3(6) can wait. Its `Magma` blocks run in seconds in the free online calculator. Everything else is by hand.

1. Integral points and Siegel's theorem

Problem 1.4 is the proof of [Sil09, Corollary IX.3.2.1].

Definition 1.1 (The v-adic distance [Sil09, §IX.2])**.** Let C be a smooth projective curve over K, let $v \in \mathcal{P}_K$, and let $Q \in C(K_v)$. Choose a function $t_Q \in K_v(C)$ that has a zero of order $e \geqslant 1$ at Q and no other zeros. The *v-adic distance* from $P \in C(K_v)$ to Q is

$$d_v(P, Q) \coloneqq \min\big(|t_Q(P)|_v^{1/e}, 1\big),$$

with $d_v(P, Q) \coloneqq 1$ if P is a pole of t_Q.

The distance $d_v(P, Q)$ is small when $P, Q \in C(K_v)$ are v-adically close.

Definition 1.2 (Heights on an elliptic curve [Sil09, §VIII.6])**.** Let E be an elliptic curve over K, and let $f \in K(E)$ be a nonconstant function. The *height relative to f* of a rational point $P \in E(K)$ is $h_f(P) \coloneqq \frac{1}{[K:\mathbb{Q}]} \log H_K(f(P))$, with $h_f(P) \coloneqq 0$ if P is a pole of f. The function f is *even* if $f(-P) = f(P)$ for every $P \in E(\overline{K})$.

With the height h_f of Definition 1.2, Siegel's theorem says that, when $E(K)$ is infinite, the rational points P of an elliptic curve E cannot approach Q quickly, compared with their height: for every $\varepsilon > 0$, all but finitely many of them satisfy $d_v(P, Q) \geqslant \exp(-\varepsilon h_f(P))$.

Theorem 1.3 (Siegel [Sil09, Theorem IX.3.1])**.** *Let E be an elliptic curve over K such that $E(K)$ is infinite. Let $Q \in E(\overline{K})$, let $f \in K(E)$ be a nonconstant even function, and let v be a place of the field $K(Q)$, with $d_v(\cdot, Q)$ defined on E as a curve over $K(Q)$. Then*

$$\lim_{\substack{P\in E(K)\\ h_f(P)\to\infty}} \frac{\log d_v(P, Q)}{h_f(P)} = 0.$$

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Problem 1.4 (Integral points on elliptic curves)**.**
Let E be an elliptic curve over K, given by a Weierstrass equation with coordinate functions x and y.

(1) Show that $t_O := 1/x$ has a zero of order 2 at O and no other zeros. Deduce that, for this choice of t_O, every $v \in \mathcal{P}_K$ and every $P \in E(K_v) \setminus \{O\}$,

$$d_v(P, O) = \max\big(1, |x(P)|_v\big)^{-1/2}.$$

(2) Let $P \in E(K) \setminus \{O\}$ be a rational point with $x(P) \in \mathcal{O}_S$. Show that

$$h_x(P) = \frac{1}{[K : \mathbb{Q}]} \sum_{v \in S} n_v \log \max\big(1, |x(P)|_v\big),$$

and that some $v \in S$ satisfies $h_x(P) \leqslant -2\,\#S \log d_v(P, O)$, with d_v as in (1).

(3) Show that the set $\{P \in E(K) \setminus \{O\} : x(P) \in \mathcal{O}_S\}$ is finite.[a]

[a] *Hint:* if not, some place $v \in S$ serves in (2) for infinitely many rational points of the set.

2. The equation $y^2 = f(x)$ and the S-unit equation

Problem 2.3 is the proof of [Sil09, Theorem IX.4.3]. It relies on the following theorem.

Theorem 2.1 (The S-unit equation [Sil09, Theorem IX.4.1])**.** *Let $\lambda, \mu \in K^\times$. Then the equation $\lambda u + \mu w = 1$ has only finitely many solutions $(u, w) \in \mathcal{O}_S^\times \times \mathcal{O}_S^\times$.*

Definition 2.2 collects Silverman's reductions, together with one he omits: the roots of the polynomial are S-integers.

Definition 2.2 (Adapted sets of places)**.** Let $f \in K[T]$ be a polynomial of degree $d \geqslant 1$. The set S is adapted to f if the following conditions hold.

(1) The polynomial f splits over K, as $f = c\prod_{i=1}^{d}(T - \alpha_i)$, with $c \in \mathcal{O}_S^\times$ and $\alpha_1, \ldots, \alpha_d \in \mathcal{O}_S$.
(2) For all $i \neq j$, $\alpha_i - \alpha_j \in \mathcal{O}_S^\times$.
(3) The ring $\mathcal{O}_S$ is a principal ideal domain.

Problem 2.3 (The equation $y^2 = f(x)$)**.**
Let $f \in K[T]$ be a polynomial of degree $d \geqslant 3$ with distinct roots in $\overline{K}$. An *S-integral solution* is a pair $(x, y) \in \mathcal{O}_S \times \mathcal{O}_S$ with $y^2 = f(x)$. The goal is to show that there are only finitely many S-integral solutions.

(1) This item reduces the problem to the case where S is adapted to f. Let K' be a finite extension of K, and S' a finite set of places of K' containing the places above S.
  (a) Show that S' contains the archimedean places of K', and that $\mathcal{O}_S \subseteq \mathcal{O}_{S'}$.
  (b) Show that K' and S' can be chosen so that S' is adapted to $f \in K'[T]$.
  (c) Deduce that it is enough to prove that there are only finitely many S-integral solutions when S is adapted to f.

(2) From now on, S is adapted to f, with c and α_i as in Definition 2.2. Let (x, y) be an S-integral solution. Show that for each i there are $b_i \in \mathcal{O}_S^\times$ and $z_i \in \mathcal{O}_S$ with $x - \alpha_i = b_i z_i^2$.[a]

(3) Let $L \subseteq \overline{K}$ be the field obtained from K by adjoining a square root of every element of $\mathcal{O}_S^\times$. Show that L is a finite extension of K. Let T be the set of places of L above S, and $\mathcal{O}_T$ the ring of T-integers of L. Choose $\beta_i \in L$ with $\beta_i^2 = b_i$. Show that for all $i \neq j$,

$$\beta_i z_i + \beta_j z_j \in \mathcal{O}_T^\times \quad \text{and} \quad \beta_i z_i - \beta_j z_j \in \mathcal{O}_T^\times.$$

(4) Let $\theta := \beta_1 z_1 - \beta_3 z_3$, which is nonzero by (3). Verify *Siegel's identities*

$$\frac{\beta_1 z_1 + \beta_2 z_2}{\theta} - \frac{\beta_2 z_2 + \beta_3 z_3}{\theta} = 1 \quad \text{and} \quad \frac{\beta_1 z_1 - \beta_2 z_2}{\theta} + \frac{\beta_2 z_2 - \beta_3 z_3}{\theta} = 1.$$

Deduce that there is a finite subset of L, depending only on K, S and f, that contains θ for every S-integral solution (x, y) and every choice of the b_i, z_i and β_i.[b]

(5) Show that there are only finitely many S-integral solutions. Where did the proof use that $d \geqslant 3$?

(6) Let $K = \mathbb{Q}$, $S = \{\infty, 2, 3\}$ and $f = T^3 - 36T = T(T-6)(T+6)$, with $\alpha_1 = 0$, $\alpha_2 = 6$ and $\alpha_3 = -6$. Here $\mathcal{O}_S = \mathbb{Z}[1/6]$.

(a) Show that S is adapted to f. Find $\mathcal{O}_S^\times$, representatives of $\mathcal{O}_S^\times / \mathcal{O}_S^{\times 2}$, and the field L.

(b) For each of the S-integral solutions $(-3, 9)$, $(12, 36)$, $(18, 72)$ and $(294, 5040)$, write $x - \alpha_k = b_k z_k^2$ with $b_k \in \{\pm 1, \pm 2, \pm 3, \pm 6\}$ and z_k a positive integer.

(c) For the S-integral solution $(12, 36)$, take $\beta_1 = \sqrt{3}$, $\beta_2 = \sqrt{6}$ and $\beta_3 = \sqrt{2}$. The code below computes the terms u and w of the first of Siegel's identities, and their minimal polynomials over $\mathbb{Q}$. Explain how the output shows that $u, w \in \mathcal{O}_T^\times$.

```
R<t> := PolynomialRing(Rationals());
L<i, r2, r3> := NumberField([t^2 + 1, t^2 - 2, t^2 - 3] : Abs);
c1 := 2*r3;  c2 := r2*r3;  c3 := 3*r2;       // beta_k*z_k for (12, 36)
u := (c1 + c2)/(c1 - c3);
w := -(c2 + c3)/(c1 - c3);
u + w;                          // 1
MinimalPolynomial(u);           // t^4 + 8*t^3 + 2*t^2 - 8*t + 1
MinimalPolynomial(w);           // t^4 - 12*t^3 + 32*t^2 - 24*t + 4
```

(d) The intrinsic `SIntegralPoints` returns all the rational points $P \neq O$ of the elliptic curve $y^2 = x^3 - 36x$ with $x(P) \in \mathbb{Z}[1/6]$, one for each pair $\pm P$. The code below lists their x-coordinates. One of them is not an integer. Find its b_k and z_k.

```
E := EllipticCurve([-36, 0]);
[ Q[1] : Q in SIntegralPoints(E, [2, 3]) ];
// [ -6, -3, -2, 0, 6, 25/4, 12, 18, 294 ]
```

[a] *Hint:* a prime element of $\mathcal{O}_S$ divides at most one of the $x - \alpha_i$.

[b] *Hint:* multiply the first terms of the two identities.

References

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑